PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA DISTRICT COUNCIL

DISTRICT FORM FOUR MOCK EXAMINATION

CODE: 033/1 **BIOLOGY 1**

TIME: 3:00 HOURS ElimuYako.co.tz **2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A, and B and two questions from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. All writings should be in blue or black pen, except for diagrams that must be drawn in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet(s).

SECTION A (16 MARKS)

Answer ALL questions from this section

1. For each of the items (i) - (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.
   (i) Maintenance of constant internal environment of a body is .....
   A. Secretion
   B. Osmoregulation
   C. Homeostasis
   D. Regulation
   E. Peristalsis

(ii) If the human blood is mixed with distilled water and examined under microscope, the red blood cells would have:

A. Expanding
B. Shrunk a little
C. Swollen slightly
D. Stayed in some size
E. Swollen and bust

(iii) A biologist discovered a new cell in a culture. The new cell had district cell wall but did not have a definite nucleus. The cell is most likely to be:

A. Animal cell
B. Fungi
C. Prokaryote
D. Protozoa
E. Eukaryote

(iv) In which of the following mammalian blood vessel does deoxygenated blood flow away from the heart?

A. Aorta
B. Vena cava
C. Ventricle
D. Pulmonary artery
E. Pulmonary vein

(v) The function of bile is to:

A. Soften food
B. Produce gall stones
C. Deaminate protein
D. Digest fat
E. Emulsify fats

(vi) Part of cassava plant which is modified for food storage is:

A. An underground stem
B. Underground root
C. A corn
D. An underground stalk
E. A fruit

(vii) All of these are characteristics of respiratory surfaces except:

A. Thin
B. Moist
C. Have large surface area
D. Far from an efficient transport system
E. Well supplied with blood capillaries

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(viii) Nerve cells are interconnected by:

A. Synapse
B. Cell body
C. Dendrites
D. Axon
E. Myelin sheath

(ix) If a parent cell has twelve chromosomes, how many chromosomes will be in the daughter cells after meiosis?

A. Six (6)
B. Twenty-four (24)
C. Eight (8)
D. Thirty-six (36)
E. Twelve (12)

(x) When a human being stretches out his or her fore arm to pick a phone from a table, the motion of the arm is brought about by:

A. Contraction of the biceps and relaxation of triceps
B. Stretching of skin and relaxation of ligaments
C. Contraction of triceps and relaxation of biceps
D. Relaxation of both biceps and triceps
E. Contraction of the flexor and extensor muscles

2. Match the phrases in List A with the responses in List B by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i. The type of growth that occurs in a series of stages ii. The type of growth whereby all body organs grow at the same rate iii. The type of growth whereby organism stop growing when a certain body size or age is attained iv. The type of growth whereby different parts of the body of an organism grow at different rates and stop growing at different times v. The type of growth whereby growth occurs all over the body of an organism vi. The type of growth whereby growth occurs in certain regions of an organism	A. Diffuse growth B. Localized growth C. Isometric growth D. Allometric growth E. Indeterminate growth F. Determinate growth G. Intermittent growth

SECTION B (54 MARKS)

Answer ALL questions from this section

3. (a) For the following situations, decide whether the muscles cells in human body are likely to be relying mainly on aerobic or anaerobic respiration to supply their energy requirements.
(i) A person is running a marathon distance of twenty-seven miles over five hours.
(ii) A child sprints sixty meters down a hallway in nine seconds.
(iii) A rabbit in a garden spots a fox and immediately jumps away.

(b) Describe two ways in which the body prevents food from entering the lungs.

4. (a) Other than the elimination of waste products, identify other four functions of urinary system.
(b) If the body is dehydrated, what do the kidneys do and what does this do to the appearance of the urine produced?

5. (a) Why waste disposal is a problem in our society? Give five points

(b) What can be done to reduce waste in our environment? Give four points

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6. (a) Discuss the process of absorption of water by root hairs. (any two point)
   (b) What does the circulatory system consist of? Give the main function of each component

7. (a) Explain how concentration of sugar in blood is maintained constant in humans.
   (b) Compare the functions of parts of the mammalian eye and camera. Give five points

8. (a) What are roles of decomposers in an ecosystem? Give three points
   (b) Give four differences between food chain and food web.

SECTION C: (30 Marks)

Answer **any two (2)** questions from this section.

9. Discuss how the knowledge of genetics can be applied in our country. Give six points.

10. Juma believe that all plants are good to human beings. You as a form four student give Juma awareness by tell him how plants are both good and bad to human beings. Give three points in each side (six points in total)

11. Before Biology session on reproduction in plants, form three students at Kwala secondary school were required to collect insect-pollinated flowers from the school environment. With six points explain how you can help them to identify the flowers they required to collect.